



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Autonomous Malware Analysis Research Highlights AI-Assisted Reverse Engineering Governance Risk

Threat Report

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TLP:CLEAR | Lost Boys Cyber | Autonomous Malware Analysis Research Highlights AI-Assisted Reverse Engineering Governance Risk - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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1. Executive Summary

SentinelOne research on frontier models and long-horizon malware analysis is worth analyst review because automated reversing can accelerate defenders and adversaries. Teams should govern sensitive sample handling, prompt-injection exposure, and validation of AI-generated malware findings.

This report was selected for the 2026-07-27 daily Lost Boys Cyber production run. Source depth was sufficient for a defender-priority report, but any victim-specific details, exploitability claims, or attribution beyond the cited sources should be validated before operational use.

2. Intelligence Requirements

Question	Current Answer	Confidence
Why prioritize now?	Recent or materially updated public reporting places this issue in the daily watch window.	Medium
Who should review first?	malware analysis, SOC automation, and AI governance defenders and asset owners.	Medium
What telemetry matters?	Identity, endpoint process, network, SaaS, edge, mail, OT, and supplier-access telemetry as applicable.	Medium

3. Source Review & Confidence

#	Source	URL	Contribution
1	SentinelOne frontier models malware analysis	https://www.sentinelone.com/labs/frontier-models-tackle-autonomous-long-horizon-malware-analysis/	Supports selection, confidence, or defender action.
2	NIST AI Risk Management Framework	https://www.nist.gov/itl/ai-risk-management-framework	Supports selection, confidence, or defender action.
3	OWASP Top 10 for LLM Applications	https://owasp.org/www-project-top-10-for-large-language-model-applications/	Supports selection, confidence, or defender action.
Source-Reliability Assessment		Value	
Overall confidence		Medium	
Source window		Preferred 24-72 hours; expanded to 7 days or authoritative standing guidance where fewer current sector-specific sources were available.	
Evidence limits		This report does not fabricate IOCs, exploit code, victim counts, or actor attribution not present in public sources.	

4. Threat / Vulnerability Details

SentinelOne research on frontier models and long-horizon malware analysis is worth analyst review because automated reversing can accelerate defenders and adversaries. Teams should govern sensitive sample handling, prompt-injection exposure, and validation of AI-generated malware findings. The defensive focus is exposure validation, compensating control review, and near-term hunt coverage rather than speculative attribution.

5. Attack Scenarios

Scenario	Preconditions	Defensive Review
Internet-facing exposure	Affected service, gateway, mail, browser, SaaS, OT, or supplier endpoint is reachable or weakly segmented.	Confirm asset inventory, patch state, WAF/edge controls, and recent authentication anomalies.
Trusted-channel abuse	Adversary uses a browser, updater, remote-access tool, SaaS integration, supplier account, or mail workflow.	Review child-process activity, consent grants, remote sessions, new services, and unusual data movement.
Post-compromise staging	Operator or malware performs discovery, credential access, archive staging, or exfiltration.	Hunt command shells, archive tools, credential-store access, unusual outbound connections, and new persistence.

6. Detection Engineering

Signal	Telemetry Source	Analytic Goal
Rare scripting or LOLBin child process from browser/mail/remote-access/updater/service parent	EDR process creation	Surface exploitation, phishing execution, trusted-channel abuse, or operator activity.
New external connection burst from scripting, browser helper, remote admin, or OT engineering workstation	Network/proxy/EDR network	Identify C2, staging, exfiltration, or unexpected supplier access.
New service, scheduled task, local admin group change, OAuth grant, or MFA reset	Endpoint, identity, SaaS audit	Surface persistence and privilege changes.

```

title: LBC Daily Priority Suspicious Trusted Channel Execution
id: lbc-tr-2026-150-trusted-channel-exec
status: experimental
description: Detects suspicious script/LOLBin execution from parent processes commonly abused in vulnerability exploitation, phishing, browser abuse, supplier tooling, or remote administration.
logsource:
  product: windows
  category: process_creation
detection:
  parent_selection:
    ParentImage|contains:
      - '\\chrome.exe'
      - '\\msedge.exe'
      - '\\outlook.exe'
      - '\\zmmailbox'
      - '\\updater'
      - '\\vpn'
      - '\\thin'
      - '\\remote'
  child_selection:
    Image|endswith:

```

```
- '\\powershell.exe'
- '\\cmd.exe'
- '\\wscript.exe'
- '\\rundll32.exe'
- '\\mshta.exe'
- '\\curl.exe'
- '\\7z.exe'
```

```
condition: parent_selection and child_selection
level: high
```

7. Threat Hunting

7.1. Hypothesis 1: suspicious execution from trusted user-facing or supplier-managed tooling

```
DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName has_any
('chrome.exe', 'msedge.exe', 'outlook.exe', 'updater.exe', 'vpn.exe', 'remote.exe', 'thinclient.exe', 'java.exe')
| where FileName in~
('powershell.exe', 'cmd.exe', 'wscript.exe', 'rundll32.exe', 'mshta.exe', 'curl.exe', 'wget.exe', '7z.exe', 'python.exe')
| project Timestamp, DeviceName, InitiatingProcessFileName, FileName, ProcessCommandLine, AccountName
```

7.2. Hypothesis 2: abnormal outbound activity from scripting or remote-access context

```
DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName has_any
('powershell.exe', 'cmd.exe', 'curl.exe', 'wget.exe', 'python.exe', 'chrome.exe', 'msedge.exe', 'vpn.exe', 'remote.exe')
| summarize connections=count(), first_seen=min(Timestamp), last_seen=max(Timestamp),
urls=make_set(RemoteUrl, 10) by DeviceName, InitiatingProcessFileName, RemoteIP, RemotePort
| where connections > 20 or array_length(urls) > 5
```

7.3. Hypothesis 3: identity or persistence changes after initial access

```
DeviceEvents
| where Timestamp > ago(14d)
| where ActionType has_any
('UserAccountAddedToLocalGroup', 'ScheduledTaskCreated', 'ServiceInstalled', 'OAuthAppConsentGranted', 'MfaRequirementSatisfied', 'TamperingAttempt')
| project Timestamp, DeviceName, ActionType, AccountName, AdditionalFields
```

8. MITRE ATT&CK Mapping

Tactic	Technique	Rationale
Initial Access	T1190 Exploit Public-Facing Application / T1566 Phishing	Relevant to vulnerability exploitation, webmail phishing, browser abuse, supplier exposure, or OT remote-access pathways.
Execution	T1059 Command and Scripting Interpreter	Common follow-on behavior for exploitation and operator-driven activity.
Persistence / Privilege	T1053 Scheduled Task, T1543 Create or Modify System Process	Used as behavior-focused hunt coverage; validate locally.
Command and Control	T1105 Ingress Tool Transfer / T1071 Web Protocols	Covers covert browser/SaaS/HTTP C2 and staging paths.
Exfiltration	T1567 Exfiltration Over Web Service	Relevant where reporting concerns SaaS, browser, ransomware, or data-theft activity.

9. Recommended Actions

Priority	Owner	Action
Critical	Asset owners	Identify exposed products, services, browser/AI surfaces, OT remote-access systems, or supplier-managed paths relevant to this report.
High	Security operations	Run the included hunts over 14-30 days and triage hits against maintenance windows and known automation.
High	Detection engineering	Convert the Sigma and KQL into platform-native analytics with allowlists for approved administrative tools.
Medium	GRC / vendor management	Confirm notification obligations, vendor MFA, and escalation contacts for supplier-dependent operations.

10. References

- [1] SentinelOne frontier models malware analysis: <https://www.sentinelone.com/labs/frontier-models-tackle-autonomous-long-horizon-malware-analysis/>
- [2] NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- [3] OWASP Top 10 for LLM Applications: <https://owasp.org/www-project-top-10-for-large-language-model-applications/>